

Machine Learning-Based Dynamic QoS Optimization and Resource Allocation in 5G Networks

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Abstract

Fifth-generation (5G) networks support heterogeneous services with different requirements for bandwidth, latency, reliability and throughput. The dynamic nature of user traffic makes static resource allocation inefficient, particularly during congestion and sudden traffic spikes. Consequently, machine learning (ML) and reinforcement learning (RL) have become important approaches for adaptive resource management, network slicing and quality-of-service (QoS) optimization. This work reviews more than 30 representative studies covering mathematical optimization, network slicing, traffic prediction, Q-learning, deep reinforcement learning, multi-agent learning, lightweight machine learning and near-real-time O-RAN resource management. The study then develops and evaluates a predictive, user-centric bandwidth-allocation framework using real 5G application traffic traces. Random Forest is used to predict near-future demand, and a blended allocation strategy combines predicted and current demand. The experimental results show that ML prediction is substantially better than a persistence baseline, while the final predictive allocation does not outperform reactive baselines on most immediate QoS metrics. The predictive approach does, however, provide a higher fairness and resource-utilization trade-off, highlighting the practical challenge of converting accurate demand forecasts into effective resource-allocation decisions.

Keywords: 5G networks, QoS optimization, resource allocation, traffic prediction, machine learning, network slicing, reinforcement learning, dynamic bandwidth allocation, fairness

1 Related Work and Literature Review

5G networks are designed to support heterogeneous services with very different performance requirements. Enhanced Mobile Broadband (eMBB) requires high data rates, Ultra-Reliable Low-Latency Communications (URLLC) emphasizes reliability and delay, while massive Machine-Type Communications (mMTC) focuses on supporting a large number of devices. These requirements make static resource-allocation policies inadequate when traffic demand changes rapidly. Recent survey studies classify machine-learning-assisted 5G resource management into areas such as traffic prediction, admission control, network slicing, resource allocation and autonomous network management [1–3]. The literature therefore shows a clear evolution from mathematical optimization and fixed allocation rules toward adaptive learning-based approaches.

1.1 Evolution of 5G Resource Allocation

Early approaches to resource allocation formulate bandwidth, spectrum, power, computing and slice-resource assignment as constrained optimization problems. These formulations can provide strong or even optimal solutions for a given network state, but their computational requirements increase rapidly as the number of users, slices, resource blocks and QoS constraints grows. This motivated learning-based approaches in which a model learns an allocation policy from network observations and performs the final decision with a relatively small online computation cost.

Network slicing became an important mechanism for isolating resources and satisfying heterogeneous service requirements. DeepSlicing combines the Alternating Direction Method of Multipliers (ADMM) with deep reinforcement learning (DRL) to allocate resources among slices [4]. Other works address dynamic slice reconfiguration and large action spaces using deep Q-learning, demonstrating that deep reinforcement learning can adapt allocations when traffic load changes [5]. Reinforcement-learning approaches have also been applied to joint frequency-time, transmit-power and processor-resource allocation while incorporating priority, latency and throughput requirements [6]. These works established DRL as an important direction for dynamic 5G resource management.

1.2 Reinforcement Learning and Deep Reinforcement Learning

In reinforcement learning (RL), an agent observes a state s_t , selects an action a_t , receives a reward r_t , and transitions to a new state s_{t+1} . Q-learning updates the estimated action value according to

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right], \quad (1)$$

where α is the learning rate and γ is the discount factor. When the state or action spaces become large, a neural network can approximate the action-value function, leading to DQN-based methods.

A large body of 5G resource-allocation research uses DQN, actor–critic methods, multi-agent reinforcement learning (MARL), or hierarchical DRL. Intelligent eMBB/URLLC slicing combines optimization with DRL to satisfy rate and reliability requirements [7]. Energy-efficient DRL has been used to jointly manage radio and power resources [9]. Other studies

consider beyond-5G slicing, model-based reinforcement learning, dynamic multi-tenant slicing, interference-aware mixed numerology allocation, and online allocation [12–15, 23]. Multi-agent learning extends the framework to settings in which several agents coordinate decisions over power, computing and radio resources [17, 36].

Although RL/DRL provides strong adaptability, it introduces its own challenges. Training can be computationally expensive, reward design can strongly influence the learned policy, and exploration may lead to unstable or undesirable decisions. These issues have motivated approaches based on transfer learning, safe reinforcement learning and hierarchical control. Deep transfer reinforcement learning has been used for joint radio and cache allocation to reduce adaptation cost [18]. Safe-and-fast RL explicitly targets safe exploration, faster convergence and data efficiency [24]. Hierarchical DRL decomposes the problem into slice-level and user-level decisions to improve scalability [21].

1.3 Traffic Prediction as a Basis for Proactive Allocation

A different research direction is to predict future traffic before making resource-allocation decisions. This changes the control strategy from reactive allocation to proactive allocation. If $D_{u,t}$ denotes the demand of user u at time t , a prediction model can estimate

$$\hat{D}_{u,t+1} = f_{\theta}(X_{u,t}), \quad (2)$$

where $X_{u,t}$ represents a window of historical traffic and network observations and f_{θ} is a learned prediction function.

Rony et al. proposed a machine-learning-based traffic-prediction mechanism followed by dynamic spectrum allocation in 5G. Their results show that prediction-assisted allocation can approach an ideal dynamic system and improve spectrum utilization and allocation accuracy relative to static approaches [8]. This work is particularly relevant to the present study because it establishes the basic idea that future demand can be predicted and used to guide resource allocation. However, its allocation perspective is primarily network/cell-level rather than a broad per-user QoS framework.

Traffic prediction and allocation have also been combined using deep neural networks. Konstantoulas et al. proposed a framework using LSTM and Transformer-based models for traffic prediction and a CNN-based module for user allocation. The study uses real 5G mobile-terminal traffic and DeepMIMO data and reports traffic-prediction accuracy above 99% in its experiments [30]. While highly relevant, the allocation task is centered on user-to-base-station assignment rather than directly optimizing a unified bandwidth-allocation objective across latency, throughput, packet loss, jitter and fairness.

Other predictive methods forecast network state, computing capacity, mobility or slice demand and subsequently adapt network resources. Staged reinforcement learning has used a neural network to estimate future resource usage before selecting slice requests [19]; LSTM combined with A2C has been applied to mobility-aware RAN slicing [20]; and predictive slice-reconfiguration methods have investigated ML-based forecasts of future computing capacity and traffic saturation. These works indicate that prediction can complement, rather than replace, the allocation controller.

1.4 User-Centric QoS and Dynamic Bandwidth Allocation

The most direct recent comparison to the proposed work is the user-centric bandwidth-allocation literature. Chia et al. proposed User Priority Resource Allocation (UPRA) and Enhanced-UPRA (EUPRA), and then integrated Q-learning and DQN so that resource allocation changes with network state. Their evaluation uses 5G simulation with 10–100 user equipments per node and reports a 12.73% QoE improvement for EUPRA-DQN over a network-slicing baseline in overloaded conditions [31]. This work demonstrates that user priority and ML-based dynamic allocation are effective under overload. However, it does not explicitly predict future per-user demand and does not use the complete multi-metric evaluation proposed in this project.

Other recent studies concentrate on end-to-end resource allocation, MEC, multi-resource slicing and QoS satisfaction. Dynamic end-to-end allocation has been studied using multi-agent deep reinforcement learning with SLA, delay and cost objectives [27]. Multi-resource DRL methods target QoS satisfaction while jointly allocating several resource types [28]. Hybrid RL architectures combine DDQN, prioritized experience replay and sequence models for service satisfaction and connectivity [29].

Near-real-time O-RAN research has further extended the problem. xSlice combines graph-based network representations with actor–critic learning to perform near-real-time resource slicing for QoS optimization [32]. This line of research is important because it moves beyond purely simulated experiments and examines resource control in a real O-RAN environment. At the same time, graph and DRL architectures introduce substantial implementation and training complexity.

1.5 Lightweight and Efficient ML-Based Allocation

Not all researchers pursue increasingly complex deep-learning architectures. Some studies investigate whether ML can approximate an optimization process while greatly reducing online execution time. The ITU-T-compliant study by Yarkina et al. evaluates polynomial regression, neural networks and random forests for 5G RAN resource arbitration [25]. The study is especially relevant to the present project because it demonstrates that lightweight supervised learning can provide a useful compromise between optimization quality and execution time.

Other work combines federated learning, mobility modelling and DRL for 5G-RAN slicing [26]. Such methods improve adaptation across multiple RAN environments but require significantly more sophisticated training and coordination.

The literature therefore suggests a practical research trade-off. Very simple heuristics have low computation but poor adaptivity. Highly advanced DRL/MARL systems are adaptive but can be expensive to train and implement. Predictive supervised ML followed by a transparent allocation policy may occupy a useful middle ground.

1.6 Multi-Objective QoS Optimization

A major challenge in resource allocation is that improving one metric can negatively affect another. Increasing the bandwidth of one user may increase its throughput but reduce resources available to other users. Similarly, maximizing throughput alone may increase delay or decrease fairness.

A general constrained bandwidth-allocation problem can be written as

$$\sum_{u=1}^U B_u \leq B_{\text{total}}, \quad (3)$$

where B_u is the bandwidth allocated to user u and B_{total} is the total available bandwidth.

A multi-objective formulation can then be written as

$$\max F = w_T T + w_Q Q + w_F F_J - w_L L - w_P P - w_J J, \quad (4)$$

where:

- T is throughput,
- Q is QoS satisfaction,
- F_J is fairness,
- L is latency,
- P is packet loss,
- J is jitter, and
- $w_T, w_Q, w_F, w_L, w_P, w_J$ are weighting factors.

Fairness can be measured using Jain's fairness index:

$$F_J = \frac{(\sum_{u=1}^U x_u)^2}{U \sum_{u=1}^U x_u^2}, \quad (5)$$

where x_u is the achieved resource or throughput for user u . The value is closer to one when the achieved allocation is more balanced.

QoS satisfaction can be represented using service requirements:

$$QoS_u = \begin{cases} 1, & T_u \geq T_u^{req}, \quad L_u \leq L_u^{max}, \quad P_u \leq P_u^{max}, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The literature contains many examples of multi-objective allocation. Some studies emphasize throughput and reliability, others focus on SLA satisfaction, energy efficiency, service satisfaction or QoE. However, these objectives are often studied separately rather than as a common evaluation framework.

1.7 Mathematical Foundation of Resource Allocation

A general resource-allocation problem can be represented using a set of users $\mathcal{U}$ and a finite resource budget. If x_u represents the fraction or amount of the resource assigned to user u , then

$$x_u \geq 0, \quad \sum_{u=1}^U x_u \leq 1. \quad (7)$$

For a demand-aware system, the ideal allocation should satisfy as much of the predicted demand as possible while preserving fairness:

$$\max_{\mathbf{x}} \sum_{u=1}^U \min(x_u, \hat{D}_u) \quad (8)$$

subject to

$$\sum_{u=1}^U x_u \leq 1. \quad (9)$$

This simplified formulation illustrates the underlying tension between resource utilization and fairness. The proposed framework extends this idea by including QoS requirements and multiple performance metrics.

1.8 Figures of Prominent Approaches

Figure 1 summarizes the evolution of intelligent 5G resource allocation. Figure 2 presents the generic predictive-QoS pipeline motivating the proposed work, while Fig. 3 summarizes the research gap identified from the reviewed studies.

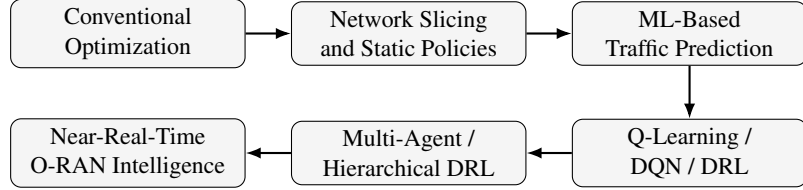


Fig. 1 Simplified evolution of major approaches for intelligent 5G resource allocation.

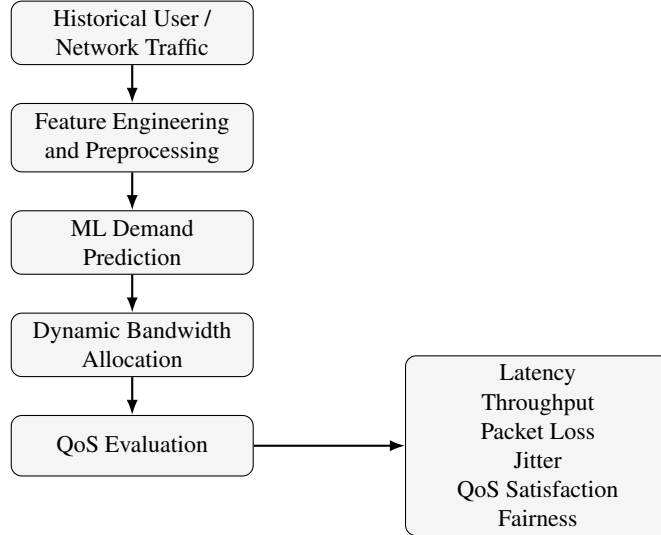


Fig. 2 Conceptual predictive QoS resource-allocation pipeline motivating the proposed research.

1.9 Complexity Analysis

Let U denote the number of users, A the number of possible actions, S the number of states, P the number of trainable model parameters, T the prediction-window length, d the hidden dimension, L the number of GNN layers, and E_g the number of graph edges.

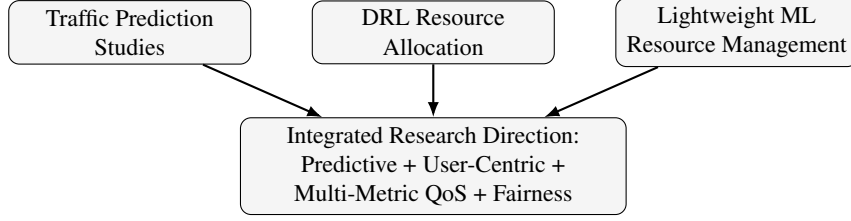


Fig. 3 Research gap identified by combining complementary directions in the literature.

Table 1 Complexity analysis of representative resource-allocation approaches.

Method	Decision Complexity	Training / Optimization	Space	Scalability Observation
Equal Share	$O(U)$	None	$O(U)$	High
Round Robin	$O(U)$	None	$O(U)$	High
Priority Allocation	$O(U \log U)$	None	$O(U)$	High-Medium
Proportional Fair	$O(U) - O(U \log U)$	None	$O(U)$	High-Medium
Exact Combinatorial Optimization	Problem-dependent; can become exponential	Potentially exponential / NP-hard	High	Poor for large dynamic problems
Tabular Q-Learning	$O(A)$	Approx. $O(N_{\text{steps}}A)$	$O(SA)$	State/action growth
DQN / DDQN	$O(P)$ forward pass	Approx. $O(BP)$ per update	$O(P)$ + replay buffer	Medium; training-heavy
Actor-Critic	$O(P_a + P_c)$	Approx. $O(B(P_a + P_c))$	$O(P_a + P_c)$	Medium
Multi-Agent DRL	$O(\sum_i P_i)$	Grows with agent count	$O(\sum_i P_i)$ + buffers	Coordination overhead
LSTM Prediction	$O(Td^2)$ per layer	Approx. $O(BTd^2)$	$O(Td^2)$	Good for moderate sequences
Transformer Prediction	$O(T^2d + Td^2)$	Same order during training	$O(T^2)$ attention memory	Cost rises for long sequences
GNN + DRL	$O(LE_g d)$ + policy cost	Higher than standalone DRL	$O(LE_g d)$ + parameters	Medium; topology-aware but complex

The complexity comparison demonstrates an important design trade-off. Rule-based policies have very low online complexity but cannot anticipate future demand. Exact optimization can achieve high-quality decisions but may become impractical as the problem size grows. DRL moves much of the computation to a training stage, but the training pipeline is more complex. Lightweight supervised prediction followed by a transparent allocation policy therefore offers a practical middle ground.

The ITU-T-compliant ML study is particularly relevant because it demonstrates that supervised ML can substantially reduce the runtime of resource arbitration relative to exact optimization in its evaluation [25].

1.10 Comparison of Existing Studies

Table 2 compares 32 representative studies using more than six parameters. The comparison considers year, learning approach, prediction, allocation level, environment or scale, QoS focus, fairness and the major limitation relevant to the proposed research.

Table 2: Comparison of representative ML/RL-based 5G QoS and resource-allocation studies.

#	Study	Year	ML / AI	Prediction	Allocation Level	Environment / Scale	Main QoS Focus	Major Limitation
1	DeepSlicing	2020	ADMM + DRL	No	Network slices	Simulation	Throughput, latency	Hybrid optimization complexity
2	Network Slice Reconfiguration with Large Action Space	2020	DQN	No	Slice resources	Simulation	Resource efficiency	Large action-space complexity
3	Dynamic Resource Optimization in 5G RAN Slicing	2020	Q-Learning	No	Frequency-time / power / compute	Simulation	Priority, throughput, latency	Reactive allocation
4	Intelligent Resource Slicing for eMBB/URLLC	2021	Optimization + DRL	No	Network slices	Simulation	Rate, reliability	Service-class specific
5	Dynamic Spectrum Allocation Following ML Traffic Predictions	2021	NARNET	Yes	Cell / spectrum	Real traffic traces	Spectrum utilization	Primarily cell-level allocation
6	Energy-Efficient DRL Resource Allocation	2021	DRL	No	Radio + power resources	Simulation	Energy efficiency	Energy-centric objective
7	Radio Resource Allocation using DRL	2021	DRL	No	RAN slices / RBs	Simulation	QoS, allocation efficiency	Limited predictive component
8	Timely Admission Control	2021	ML	Partial	Slice admission	Offline + online	Admission performance	Admission rather than detailed bandwidth allocation
9	DRL Slicing for Beyond 5G	2022	DRL	No	eMBB/URLLC/ mMTC	Simulation	Throughput, QoS	Long-term throughput emphasis
10	Model-Based RL with Kernels	2022	Model-based RL	No	RAN slices	Simulation	Learning efficiency	Sample-efficiency focus
11	Dynamic Multi-Tenant Slicing	2022	Dueling DQN	No	Multi-tenant slices	Simulation	Cost, runtime, robustness	Energy/cost and preemption focus
12	Mixed-Numerology Slice Allocation	2022	RL	No	Spectrum / slices	Simulation	Interference, reliability	PHY-specific assumptions
13	Dynamic SDN-Based RAN Slicing	2022	Multi-agent DQL / EXP3	No	gNB and users	Simulation	QoS, resource efficiency	Multi-level controller complexity
14	NOMA-Enabled MEC Allocation	2022	Multi-agent RL	No	Power + computing	NOMA/MEC simulation	Effective throughput	NOMA-specific setting
15	Deep Transfer RL for Radio/Cache	2022	DTRL	No	Radio + cache	RAN slicing simulation	Delay, throughput	Transfer architecture complexity
16	Staged 5G/B5G Slice Management	2023	DNN + DRL	Yes	Slice requests	Simulation	Slice response / resource use	Prediction is slice-oriented

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Table 2 (continued)

#	Study	Year	ML / AI	Prediction	Allocation Level	Environment / Scale	Main QoS Focus	Major Limitation
17	Efficient Dynamic RAN Slicing	2023	LSTM + A2C	Partial	RAN slicing / mobility	Mobility simulation	Service satisfaction, spectrum efficiency	Mobility-focused prediction
18	Hierarchical Scalable DRL	2023	Hierarchical TD3	No	Slice + user agents	Example: 3 slices / 15 users	QoS, utility, fairness	Comparatively small demonstration scale
19	AI Allocation under Demand/CSI Uncertainty	2023	DRL	Partial	End-to-end slicing	Simulation	Utility, resource allocation	Uncertainty rather than explicit forecasting
20	Online Resource Allocation in Slicing	2023	DRL	No	RAN slices	Simulation	QoS / resource allocation	Online learning complexity
21	Safe and Fast RL	2023	Safe RL	No	RAN slicing	Simulation	SLA success, convergence	Limited full QoS dashboard
22	ITU-T-Compliant ML Enhancement	2024	Polynomial / NN / RF	No	Slice resource arbitration	Traffic + mmWave	Accuracy, time, fairness, SLA	High-load SLA issues
23	Mobility-Aware Federated DRL	2024	SBiLSTM + Fed-A2C	Partial	Multi-RAN / users	Multi-RAN simulation	Energy, convergence, isolation	High architectural complexity
24	Dynamic End-to-End MADRL Allocation	2024	Multi-agent DRL	Partial	End-to-end + MEC/NFV	Dense-user simulation	SLA, cost, delay	Coordination complexity
25	Hybrid RL for 5G Slicing	2025	DDQN-PER + LSTM/PtrNet	Partial	RAN slicing	Multiple user scenarios	QoE, satisfaction, efficiency	Hybrid architecture complexity
26	Konstantoulas et al.	2025	LSTM / Transformer + CNN	Yes	User-to-BS allocation	Real traffic + DeepMIMO	Prediction, allocation	Not a full bandwidth/QoS optimizer
27	Chia et al.	2025	Q-Learning + DQN	No	User bandwidth	10–100 UEs/node	QoE, throughput, delay, jitter	No explicit future-demand prediction
28	xSlice	2025/26	Actor-Critic + GCN	No explicit forecast	Near-RT O-RAN	10 smartphones	Throughput, latency, reliability	High implementation complexity
29	Joint QoS and Energy-Efficient Allocation	2023	RF + Gradient Boosting	Partial	User / slice resources	Simulation	QoS, energy	Not a full predictive QoS pipeline
30	ML Forecasting for Dynamic Slice Reconfiguration	2023	ML forecasting	Yes	Virtualized computing resources	Simulation	Forecasting, saturation avoidance	Narrower resource dimension
31	Multi-Resource Network Slicing with DRL	2024	DRL	No	Multi-resource slices	Simulation	QoS satisfaction	Narrower metric set
32	Twin-GAN / DRL Hybrid C-RAN	2023	GAN + DRL	Partial	Wireless + computing	C-RAN simulation	Resource efficiency	High model complexity

The comparison demonstrates that existing research is already mature in ML-based resource allocation. However, the approaches generally specialize in one or two dimensions, such as traffic prediction, throughput maximization, energy efficiency, service satisfaction, reliability or dynamic slicing. This provides the motivation for an integrated predictive user-centric framework.

1.11 Quantification of Literature Limitations

The reviewed literature was also examined quantitatively in order to avoid making only qualitative claims about existing research.

Among the 32 studies reviewed in this work, approximately 25 employ reinforcement learning or deep reinforcement learning as a primary resource-allocation mechanism. This corresponds to

$$\frac{25}{32} \times 100 = 78.1\%. \quad (10)$$

Thus, the selected literature set is strongly concentrated around reinforcement-learning-based control.

A smaller subset explicitly predicts future traffic or future resource demand and then uses the prediction in a subsequent allocation decision. At least five studies in the screened set contain a clear predictive component of this type. This indicates that proactive prediction-driven allocation is less common than direct reactive learning-based allocation.

Real physical or near-real deployment is also uncommon. One study in the reviewed set provides a clearly documented physical O-RAN testbed evaluation as a major component of its resource-allocation experiment. This corresponds to

$$\frac{1}{32} \times 100 = 3.1\%. \quad (11)$$

Explicit jitter evaluation is also uncommon. The 2025 user-bandwidth study is particularly notable for directly considering jitter together with throughput and delay [31].

Most importantly, no paper in the reviewed set was identified as jointly using the complete evaluation tuple

$$\{\text{latency, throughput, packet loss, jitter, QoS satisfaction, Jain fairness}\} \quad (12)$$

as a unified allocation-evaluation framework.

These percentages are intentionally limited to the 32 papers selected for this review and are not presented as universal statistics for the entire 5G literature. Their purpose is to quantify the distribution of approaches within the reviewed literature.

1.12 Research Gap and Positioning of the Proposed Work

The literature demonstrates that machine learning for 5G resource allocation is already a mature research direction. Therefore, the novelty of the proposed work is not simply the use of machine learning.

Instead, the research is positioned around the integration of four ideas:

1. prediction of near-future traffic demand at the user level,

2. dynamic bandwidth distribution using predicted demand,
3. simultaneous consideration of multiple QoS requirements, and
4. explicit evaluation of fairness together with QoS.

The core conceptual flow is

$$\begin{aligned} &\text{historical traffic} \rightarrow \text{ML demand prediction} \\ &\rightarrow \text{dynamic bandwidth allocation} \rightarrow \{\text{latency, throughput, packet loss,} \\ &\quad \text{jitter, QoS, fairness}\}. \end{aligned} \quad (13)$$

The central research question is therefore:

Can future user traffic demand be predicted accurately enough to support a comparatively lightweight dynamic bandwidth-allocation mechanism that improves the overall QoS–fairness trade-off under changing and overloaded 5G traffic conditions?

The proposed work is therefore intended to be evaluated against equal sharing, round robin, priority-based allocation and proportional fairness, together with selected learning-based baselines where feasible.

2 Proposed Work

The proposed work develops a machine-learning-assisted dynamic resource- allocation framework for 5G traffic. The central idea is to predict the near-future demand of each traffic stream and then use that prediction along with the currently observed demand to allocate a limited bandwidth resource. The final implementation uses a blended demand estimate rather than relying entirely on the prediction. This design was selected after validation experiments showed that combining current and future demand is more robust than using the prediction alone.

The completed framework contains four main stages: (i) traffic-data preparation, (ii) future-demand prediction, (iii) predictive-reactive bandwidth allocation, and (iv) multi-metric performance evaluation. The implemented workflow is

$$\begin{aligned} &\text{Historical Traffic} \rightarrow \text{Feature Engineering} \rightarrow \text{ML Demand Prediction} \\ &\rightarrow \text{Blended Demand} \rightarrow \text{Dynamic Bandwidth Allocation} \\ &\rightarrow \text{QoS and Fairness Evaluation}. \end{aligned} \quad (14)$$

The contribution of the work is therefore not the invention of a new learning architecture. Instead, it investigates the practical value of combining short-term demand prediction with current network-state information in a transparent bandwidth-allocation framework. The implementation is intentionally lightweight and uses Random Forest as the primary predictor so that the prediction stage remains understandable and reproducible.

2.1 Methodology

2.1.1 Traffic Dataset and Preprocessing

The experiments use the downloaded representation of a public 5G traffic collection containing application-level downstream and upstream bitrate traces. After removing the byte-identical duplicate Naver NOW file, the working dataset contains 15 unique application traffic streams and 1,182,637 rows. The streams correspond to applications represented in the source collection, including video, conferencing, gaming and metaverse traffic [30].

Each input CSV contains the fields `DL_bitrate` and `UL_bitrate`. The downloaded representation does not document a wall-clock timestamp or a definitive bitrate unit. Consequently, the implementation preserves the original values, uses row order as a sequential observation index, and does not claim an undocumented Mbps or Kbps conversion.

For each stream, downstream bitrate is used as the primary demand signal, while upstream bitrate is retained as an additional feature. Duplicate, invalid and missing records are handled during preprocessing. One stream containing missing upstream values has the corresponding invalid rows removed while retaining the original sequential indexing information.

2.1.2 Feature Engineering

The prediction problem is constructed as a next-observation regression task. For stream u at observation t , the feature vector contains recent downstream and upstream observations and simple rolling statistics. A representative feature vector is

$$X_{u,t} = [D_{u,t}, D_{u,t-1}, D_{u,t-2}, D_{u,t-3}, D_{u,t-5}, MA_{u,t}, \sigma_{u,t}, R_{u,t}, U_{u,t}, U_{u,t-1}, U_{u,t-2}], \quad (15)$$

where D denotes downstream demand, U denotes upstream bitrate, MA is the moving average, σ is recent traffic variation, and R is the recent demand change.

Chronological splitting is used so that future observations are not randomly mixed with training data. The data is divided into training, validation and test portions, and preprocessing parameters are fitted using only the training portion.

2.1.3 Demand Prediction

Random Forest Regression is used as the primary demand-prediction model. The final configuration uses 150 trees, a maximum depth of 19 and a fixed random seed for reproducibility. A persistence baseline is also evaluated, where the next demand is assumed to equal the current demand.

The prediction target is

$$\hat{D}_{u,t+1} = f_{\theta}(X_{u,t}). \quad (16)$$

Prediction quality is evaluated with MAE, RMSE and R^2 .

2.1.4 Blended Predictive Demand

The main allocation method does not rely entirely on the predicted value. Instead, it combines the current observed demand and predicted next-step demand:

$$D_{u,t}^* = \lambda \hat{D}_{u,t+1} + (1 - \lambda) D_{u,t}, \quad (17)$$

where λ determines the contribution of prediction.

The predefined validation candidates were

$$\lambda \in \{0.25, 0.50, 0.75\}.$$

The validation MAE values were 136950.55, 115352.29 and 95543.29, respectively. Therefore, $\lambda = 0.75$ was frozen for the final test experiments. The $\lambda = 1$ configuration was retained only as a prediction-only endpoint comparison and was not used for selecting the final weight.

2.1.5 Dynamic Bandwidth Allocation

Let B_{total} denote the available bandwidth and B_u the allocation of stream u . The allocation satisfies

$$\sum_{u=1}^U B_u \leq B_{\text{total}}. \quad (18)$$

A weighted demand score combines the normalized blended demand with priority and QoS urgency:

$$W_u = \alpha \widetilde{D}_u^* + \beta P_u + \gamma Q_u, \quad (19)$$

with

$$\alpha + \beta + \gamma = 1, \quad \alpha, \beta, \gamma \geq 0. \quad (20)$$

Initial bandwidth is allocated proportionally to W_u and then capped by the blended demand. A small minimum grant is retained for active streams. Historical fairness weighting is bounded to a factor between 0.80 and 1.25 so that fairness adjustment cannot force an allocation toward zero.

2.1.6 Resource-Allocation Simulation

The completed implementation uses a Python-based resource-allocation simulation rather than a physical 5G testbed or an ns-3/5G-LENA network. The simulator uses finite queues and counts buffer overflow as packet loss. Queueing delay is derived from backlog and allocated bandwidth, subject to a 500 ms upper bound. These queue, capacity and delay values are simulation assumptions and are not properties directly measured in the source traffic dataset.

The allocation strategies compared are Equal Share, Round Robin, Proportional Fair and the proposed ML Predictive Allocation. The experiments cover 10, 25, 50 and 100 concurrent traffic streams. The 10-stream experiments use distinct real application streams. The 25-, 50- and 100-stream scenarios retain the real streams and add controlled synthetic streams derived independently from the real traffic profiles. They are therefore treated as scalability scenarios rather than measurements of 25, 50 or 100 independent physical users.

Four traffic conditions are evaluated: LOW, MEDIUM, HIGH and SPIKE. Each configuration is repeated with five seeds: 42, 43, 44, 45 and 46. This produces 320 baseline-method runs and 400 ablation runs.

2.2 Algorithm

The final algorithm is summarized in Algorithm 1. It uses the trained predictor to estimate the next demand, combines that forecast with current demand, and allocates the available bandwidth while respecting capacity, demand and fairness constraints.

Algorithm 1 Blended Predictive Dynamic QoS-Aware Bandwidth Allocation

Require: Current demand $D_{u,t}$, history $X_{u,t}$, total bandwidth B_{total}

Ensure: Allocation B_u for every active stream

- 1: Predict next demand: $\hat{D}_{u,t+1} = f_\theta(X_{u,t})$
 - 2: Compute blended demand using Eq. (17)
 - 3: Normalize blended demand
 - 4: Obtain priority and QoS-urgency terms
 - 5: Compute allocation score W_u
 - 6: Allocate B_u proportional to W_u
 - 7: Enforce $B_u \geq B_u^{\min}$ where required
 - 8: Enforce $B_u \leq D_{u,t}^*$
 - 9: Redistribute unused capacity to unsatisfied streams
 - 10: Apply bounded fairness adjustment
 - 11: Enforce $\sum_u B_u \leq B_{\text{total}}$
 - 12: Simulate finite-queue service and record QoS metrics
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2.3 Mathematical Model

Let $\mathcal{U} = \{1, 2, \dots, U\}$ denote the active traffic streams. For stream u , D_u denotes current demand, $\hat{D}_u$ denotes predicted demand, D_u^* denotes blended demand, and B_u denotes allocated bandwidth.

The predictive-reactive demand model is

$$D_u^* = \lambda \hat{D}_u + (1 - \lambda) D_u. \quad (21)$$

The allocation must satisfy

$$0 \leq B_u \leq D_u^*, \quad \sum_{u=1}^U B_u \leq B_{\text{total}}. \quad (22)$$

A multi-objective evaluation function is defined conceptually as

$$\max F = w_T \tilde{T} + w_Q \tilde{Q} + w_F F_J - w_L \tilde{L} - w_P \tilde{P} - w_J \tilde{J}, \quad (23)$$

where throughput T , QoS satisfaction Q and fairness F_J are maximized, while latency L , packet loss P and jitter J are minimized.

QoS satisfaction is measured as

$$QSR = \frac{N_{\text{satisfied}}}{U} \times 100. \quad (24)$$

Jain's fairness index is

$$F_J = \frac{(\sum_{u=1}^U x_u)^2}{U \sum_{u=1}^U x_u^2}. \quad (25)$$

The implementation records throughput, latency, packet loss, jitter, QoS satisfaction, fairness and resource utilization for every scenario.

2.4 Datasets

The working dataset contains 15 unique application-level 5G traffic streams after removal of a byte-identical duplicate Naver NOW file. The complete preprocessed collection contains 1,182,637 rows. Each input file contains downstream and upstream bitrate values. Since the available representation does not document the wall-clock observation interval or bitrate unit, the experiments preserve the supplied numeric values and use the row order as a sequential observation index.

The downstream field, `DL_bitrate`, is the primary prediction and allocation demand. `UL_bitrate` is retained as an input feature. This design avoids inventing a timestamp or applying an unsupported unit conversion.

For scalability, the 10-stream scenario uses 10 distinct real application traces. The 25-, 50- and 100-stream experiments augment the 15 real streams with synthetic streams derived independently from those real profiles. The scenario manifest records the source stream and whether each generated stream is synthetic.

The prediction stage uses a chronological train/validation/test split. The final test predictions are held out from model fitting and from selection of λ . This separation is important because the main research question concerns whether future-demand information improves allocation rather than whether the allocation algorithm can use future labels directly.

2.5 Summary of Proposed Work

The implemented system therefore follows

$$\begin{aligned} & \text{5G Traffic Streams} \rightarrow \text{Historical Features} \rightarrow \text{Random Forest Prediction} \\ & \rightarrow \text{Blended Demand} \rightarrow \text{Bandwidth Allocation} \rightarrow \text{QoS/Fairness Evaluation}. \end{aligned} \quad (26)$$

The experiments test whether accurate prediction can be converted into better resource-allocation decisions. The final results show that this conversion is not automatic: prediction accuracy improves substantially, but reactive baselines still provide stronger immediate QoS under the current simulation model. This finding is carried into the discussion in Section 3 rather than being treated as an assumption of the method.

3 Results and Discussion

The completed evaluation examines both the traffic-demand predictor and the subsequent resource-allocation strategy. The experiments cover the 15 real application traffic streams, controlled 25-, 50- and 100-stream scalability scenarios, four traffic conditions (LOW, MEDIUM, HIGH and SPIKE), and five independent random seeds. The baseline methods are Equal Share,

Round Robin and Proportional Fair. The main proposed method uses the blended demand formulation with $\lambda = 0.75$.

The results are divided into prediction accuracy, blending-weight selection, aggregate resource-allocation performance, ablation analysis and statistical testing. Figures 11–18 are generated directly from the final experiment CSV files and use the same five-seed evaluation described in Section 2. The latency, packet-loss, jitter, QoS, fairness and utilization values are generated by the Python resource- allocation simulation and therefore represent model-based evaluation rather than measurements from a physical 5G testbed.

3.1 Evaluation Parameters and Mathematical Equations

Mean Absolute Error is

$$MAE = \frac{1}{N} \sum_{i=1}^N |D_i - \hat{D}_i|. \quad (27)$$

Root Mean Squared Error is

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (D_i - \hat{D}_i)^2}. \quad (28)$$

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^N (D_i - \hat{D}_i)^2}{\sum_{i=1}^N (D_i - \bar{D})^2}. \quad (29)$$

Throughput is computed from successfully served traffic as

$$T = \frac{8D_{\text{received}}}{\Delta t}. \quad (30)$$

Average latency is

$$L = \frac{1}{N} \sum_{i=1}^N (t_i^{\text{receive}} - t_i^{\text{send}}). \quad (31)$$

Packet-loss rate is

$$PL = \frac{N_{\text{sent}} - N_{\text{received}}}{N_{\text{sent}}} \times 100. \quad (32)$$

Jitter is evaluated as the mean absolute change between consecutive simulated delays:

$$J = \frac{1}{N-1} \sum_{i=2}^N |d_i - d_{i-1}|. \quad (33)$$

QoS satisfaction is

$$QSR = \frac{N_{\text{satisfied}}}{U} \times 100. \quad (34)$$

Jain's fairness index is

$$F_J = \frac{(\sum_{u=1}^U x_u)^2}{U \sum_{u=1}^U x_u^2}. \quad (35)$$

Resource utilization is

$$RU = \frac{B_{\text{allocated}}}{B_{\text{total}}} \times 100. \quad (36)$$

Higher throughput, QoS satisfaction, fairness and utilization are preferred, while lower latency, packet loss and jitter are preferred.

3.2 Traffic-Demand Prediction Results

Table 3 compares Random Forest with the persistence baseline on the held-out prediction task.

Table 3 Held-out traffic-demand prediction performance.

Model	MAE	RMSE	R^2
Persistence	171146.79	500985.14	-0.0116
Random Forest	95939.54	275499.87	0.6941

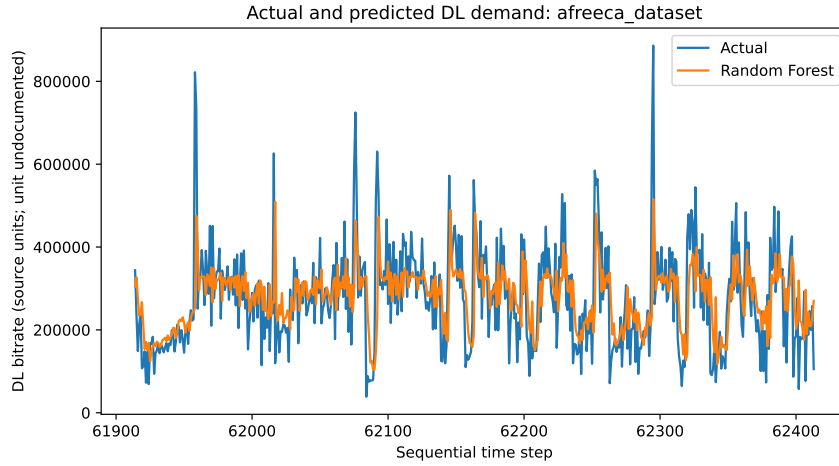


Fig. 4 Actual and Random Forest predicted traffic demand on the held-out test data.

Random Forest substantially outperforms the persistence baseline. MAE falls from 171146.79 to 95939.54, corresponding to an approximately 43.9% reduction. RMSE falls from 500985.14 to 275499.87, corresponding to an approximately 45.0% reduction. The persistence baseline has a slightly negative R^2 , whereas Random Forest achieves $R^2 = 0.6941$.

Figure 4 shows that Random Forest follows the main demand pattern, while Fig. 5 shows that the residual errors remain concentrated around periods of stronger variation. Figures 6

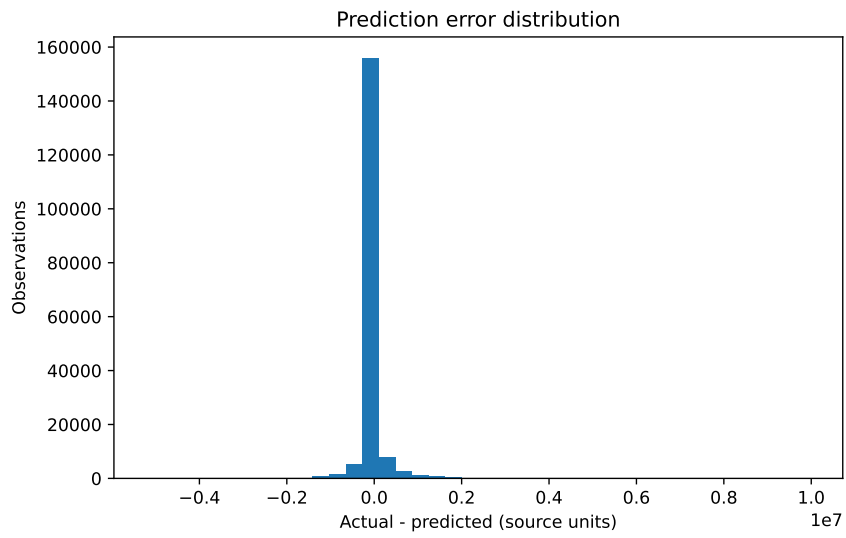


Fig. 5 Distribution of demand-prediction errors produced by the Random Forest model.

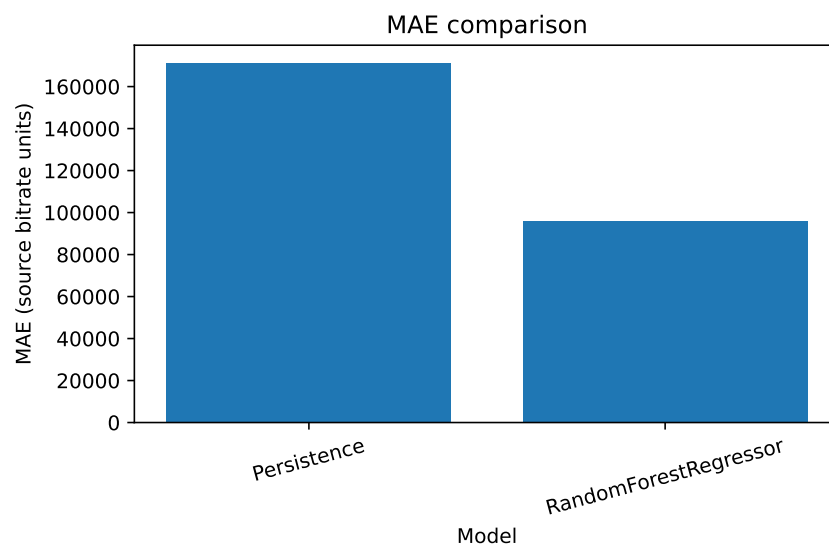


Fig. 6 MAE comparison between the persistence baseline and Random Forest prediction.

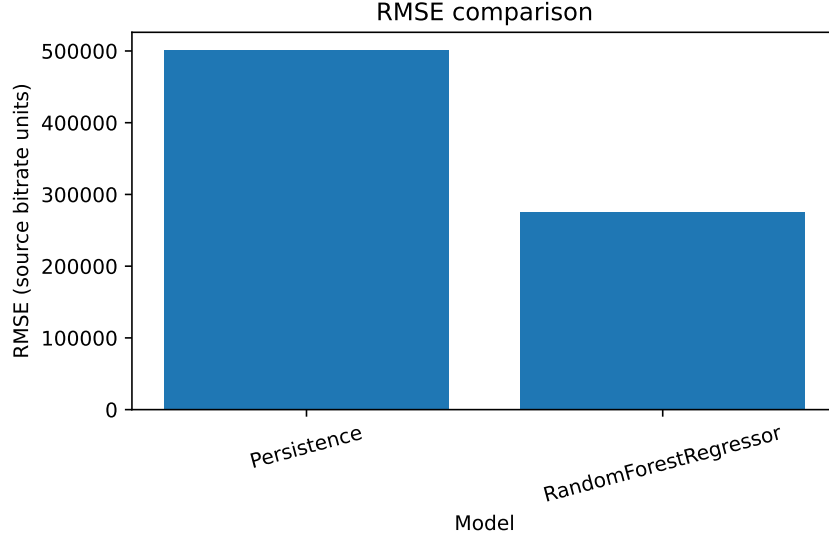


Fig. 7 RMSE comparison between the persistence baseline and Random Forest prediction.

and 7 summarize the corresponding error reductions. These results confirm that the historical traffic features contain useful information about the next demand value. However, the subsequent allocation experiments show that prediction accuracy does not by itself guarantee improved network-level performance.

3.3 Selection of the Blending Weight

The final allocator uses

$$D_{u,t}^* = \lambda \hat{D}_{u,t+1} + (1 - \lambda) D_{u,t}. \quad (37)$$

The predefined validation candidates were $\lambda = 0.25, 0.50$ and 0.75 . Their validation MAE values were 136950.55, 115352.29 and 95543.29, respectively.

Table 4 Validation selection of the demand-blending weight.

λ	Validation MAE
0.25	136950.55
0.50	115352.29
0.75	95543.29

As shown in Figure 8, validation error decreases as the prediction contribution increases across the predefined candidates. Thus, $\lambda = 0.75$ was selected using validation data only. The $\lambda = 1$ prediction-only configuration was kept as an endpoint comparison and was not used as a tuning candidate.

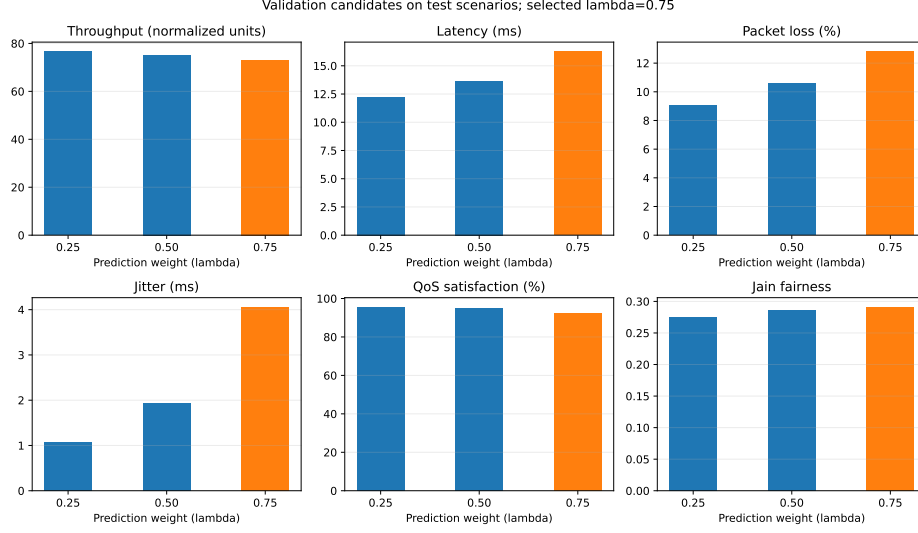


Fig. 8 Validation comparison of the candidate demand-blending weights.

3.4 Aggregate Resource-Allocation Results

The aggregate test results are shown in Table ?? . The values are means across the experimental conditions and seeds used in the completed run.

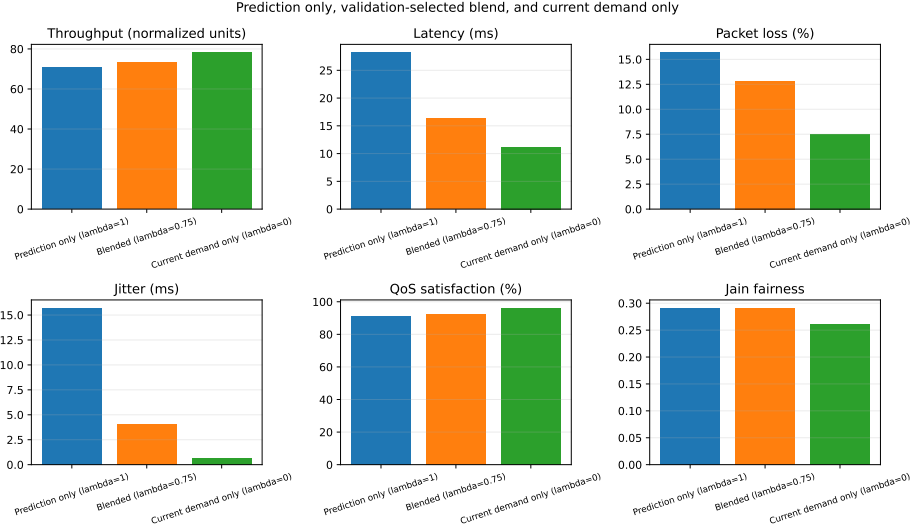


Fig. 9 Comparison of prediction-only, blended predictive, and current-demand-only allocation strategies.

Figure 9 contrasts the three demand formulations, while Figure 10 compares the proposed method with the main baselines. The current-demand-only configuration achieves the highest throughput at 78.36, followed closely by Round Robin at 78.29 and Proportional Fair at

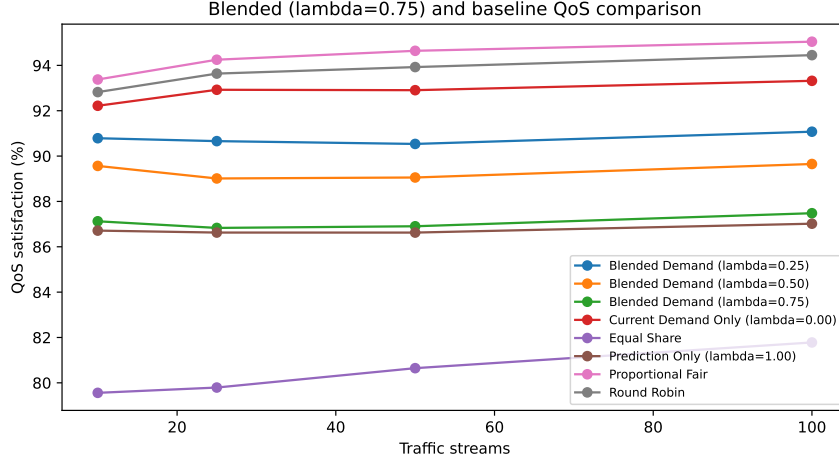


Fig. 10 Overall comparison of the proposed blended allocation method with the main baseline strategies.

78.27. The proposed blend achieves 73.15. Thus, the predictive component does not maximize immediate aggregate throughput in the current simulation.

A similar result occurs for latency. The proposed blend records 16.28 ms, compared with 10.97 ms for Round Robin and 11.12 ms for Proportional Fair. Packet loss is 12.82% for the blend, compared with 7.60% and 7.62% for Round Robin and Proportional Fair. QoS satisfaction is 92.18% for the proposed blend, again below the two reactive baselines.

The strongest positive difference appears in fairness and utilization. The blend reaches a Jain fairness value of 0.291 and resource utilization of 83.43%. Round Robin and Proportional Fair have fairness values of 0.260 and 0.267, respectively, and utilization values of 78.29% and 78.27%. Therefore, the proposed method trades some immediate QoS performance for greater resource use and a somewhat more balanced allocation.

Compared with prediction-only allocation, the blended method clearly improves the operating point. Throughput increases from 70.68 to 73.15, latency falls from 28.26 to 16.28 ms, packet loss falls from 15.72% to 12.82%, jitter falls from 15.73 to 4.06 ms, and QoS satisfaction rises from 91.15% to 92.18%. This shows that incorporating current demand reduces the negative effect of prediction error.

3.5 Evaluation Under Different Stream Loads and Conditions

Figures 11–17 show the trend of each main network metric as traffic-stream concurrency increases, while Figure 18 focuses on the sudden-demand condition. The experiments cover 10, 25, 50 and 100 streams and the four traffic conditions LOW, MEDIUM, HIGH and SPIKE. The 10-stream scenario contains distinct real application streams. The larger scenarios retain the real streams and introduce controlled synthetic streams derived from real profiles. They therefore evaluate scalability under controlled concurrency rather than representing 25, 50 or 100 independently measured physical users.

The traffic conditions are designed to increase contention from low load toward overload and sudden traffic changes. This is particularly relevant to the predictive allocation hypothesis

because prediction should be most useful when future demand changes faster than a reactive policy can respond. The experiment set contains 320 baseline-method runs and 400 ablation runs, repeated with five seeds.

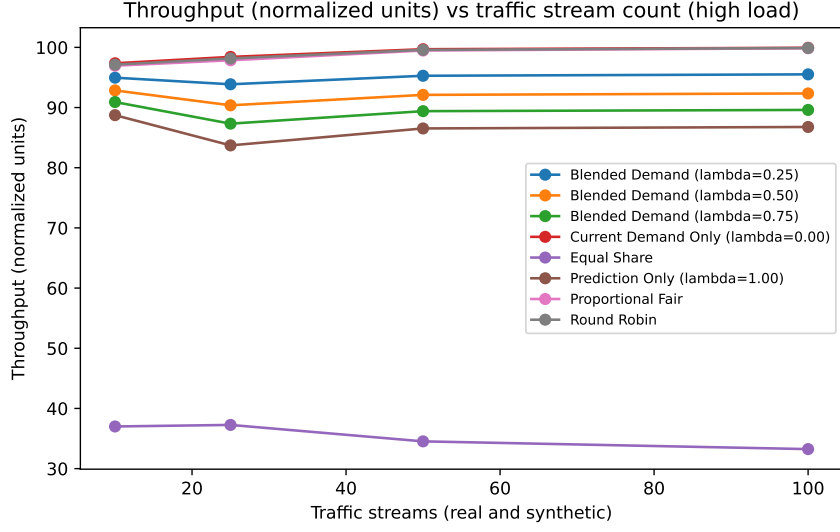


Fig. 11 Average throughput as the number of concurrent traffic streams increases.

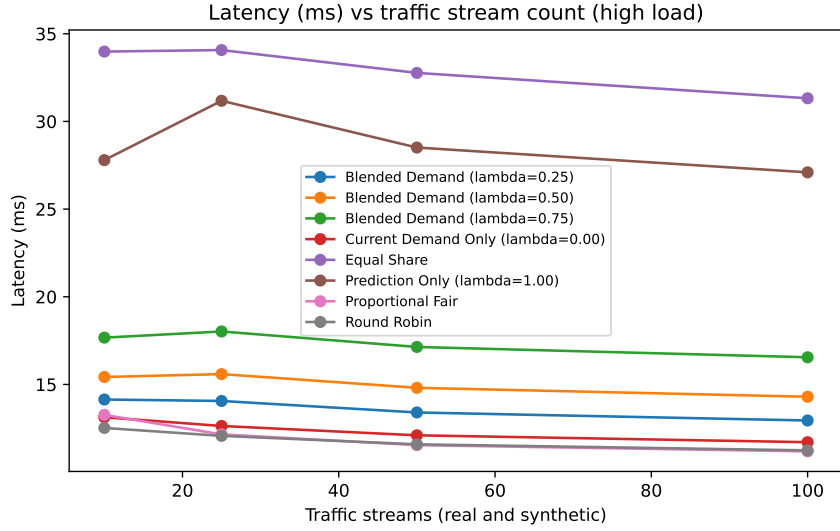


Fig. 12 Average latency as the number of concurrent traffic streams increases.

The results indicate that the proposed method remains bounded after the finite-queue and delay controls introduced in the simulator. The bounded queue model prevents the pathological

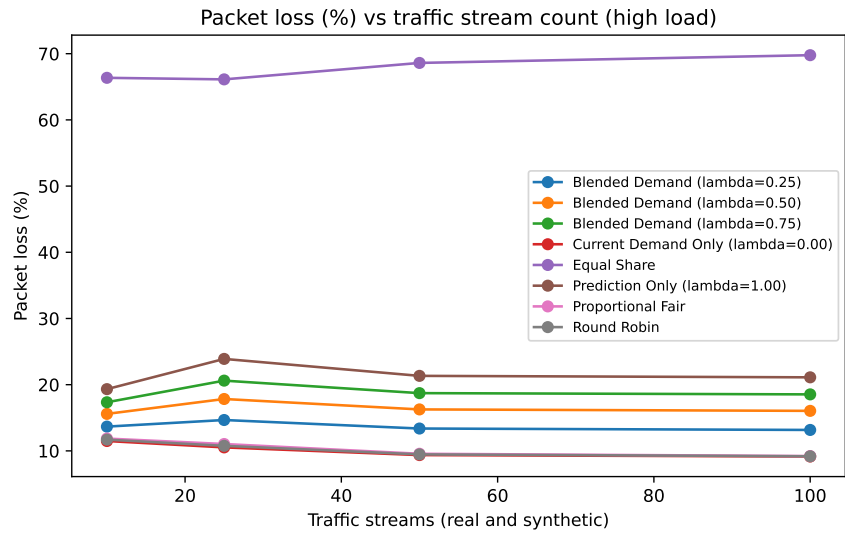


Fig. 13 Packet-loss rate under increasing traffic-stream concurrency.

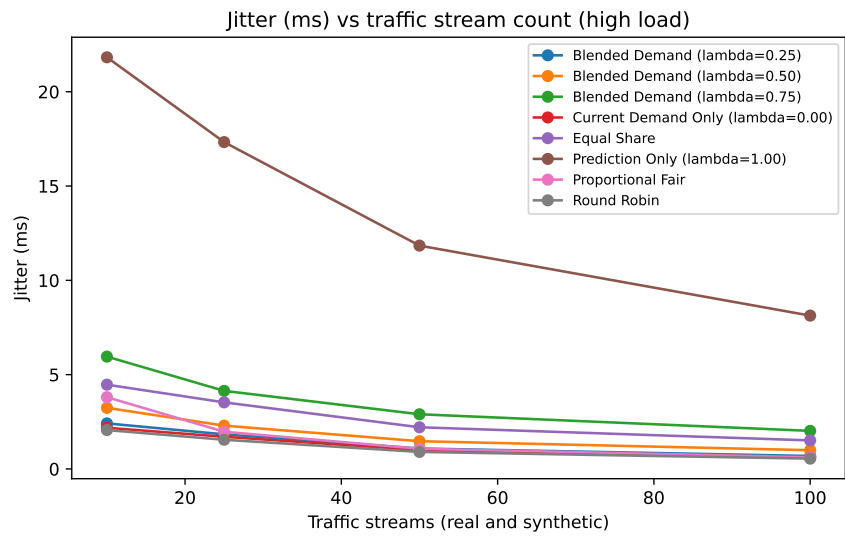


Fig. 14 Jitter under increasing traffic-stream concurrency.

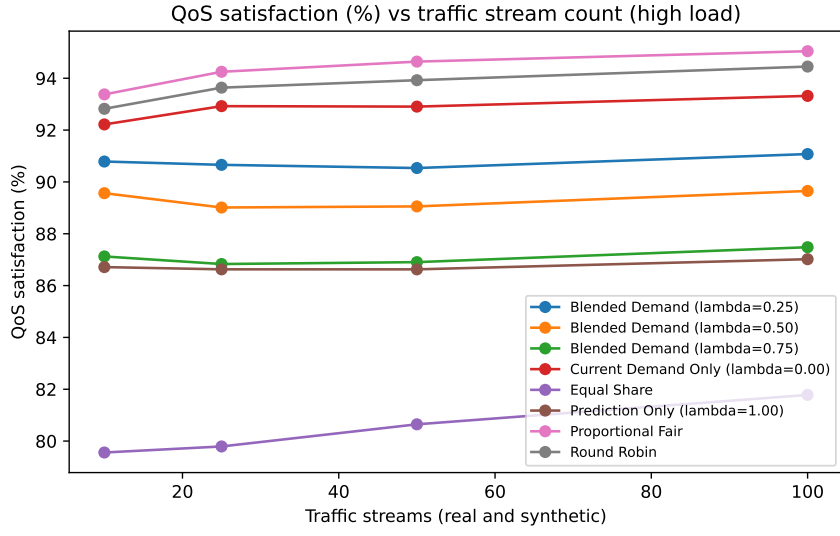


Fig. 15 QoS satisfaction ratio for increasing numbers of traffic streams.

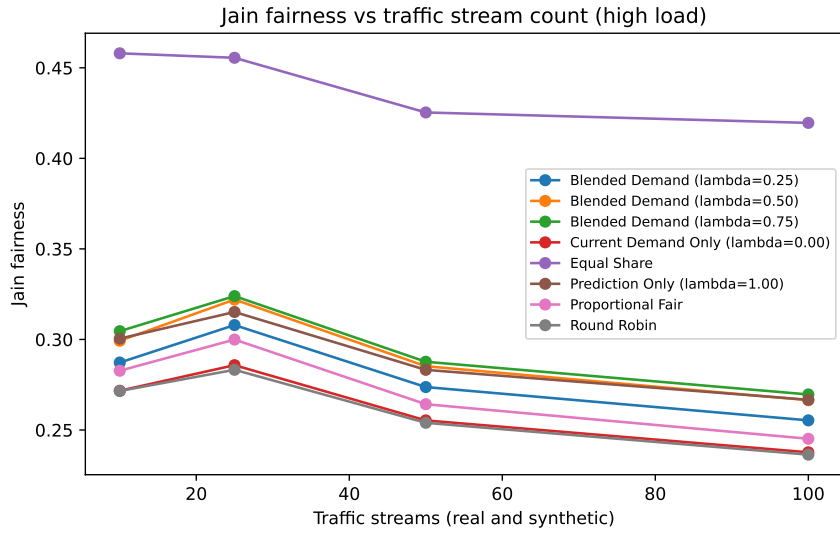


Fig. 16 Jain's fairness index under increasing traffic-stream concurrency.

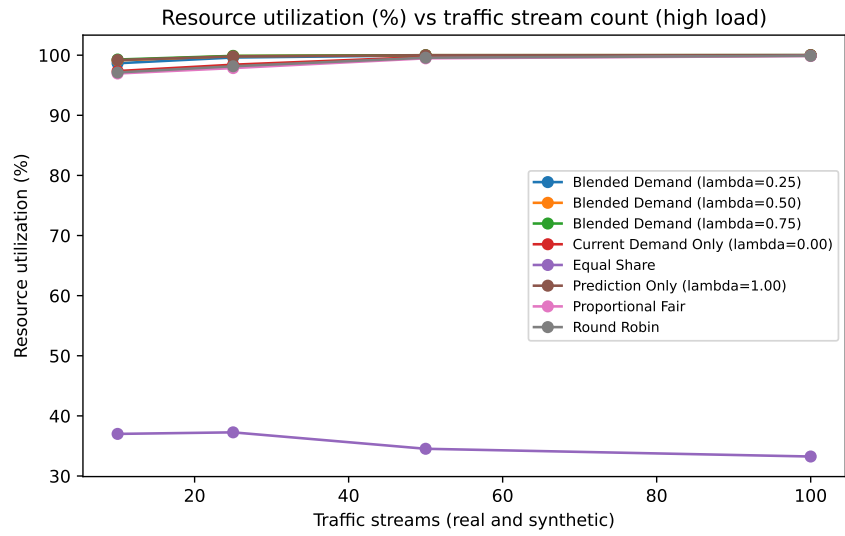


Fig. 17 Resource utilization under increasing traffic-stream concurrency.

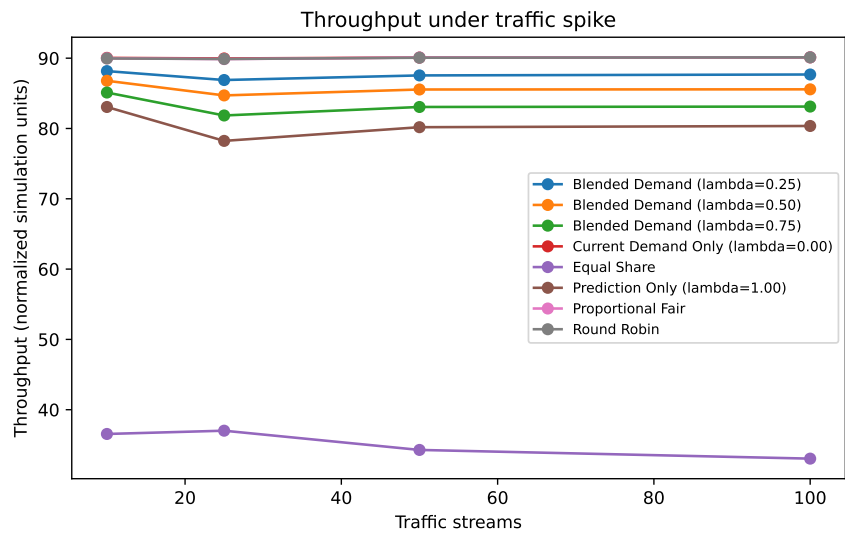


Fig. 18 Allocation behavior under a sudden traffic-spike condition.

unbounded delays seen in the initial implementation. Nevertheless, the predictive method still does not dominate reactive allocation under the aggregate evaluation.

Because all traffic traces originate from the same published collection, the experiment should not be described as a multi-dataset generalization study. It is better described as evaluation across multiple heterogeneous application traces and controlled scalability scenarios derived from those traces.

3.6 Ablation Study

The ablation study evaluates the role of prediction, fairness, priority and the multi-objective formulation. The mean results are summarized in Table ??.

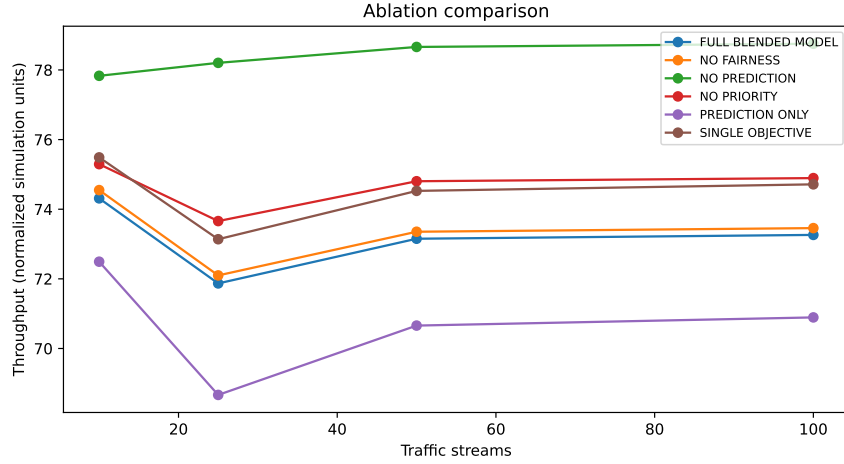


Fig. 19 Ablation comparison of the full blended model and its component-removal variants.

Figure 19 provides the visual comparison of the ablation variants. The most important ablation is the comparison between the full model and No Prediction. Removing prediction improves throughput, latency, packet loss, jitter and QoS satisfaction, while reducing fairness. This means that prediction accuracy alone does not guarantee improved allocation. The prediction error is still large enough that using the forecast as an allocation signal can hurt immediate network performance.

The fairness component itself produces a small fairness change: the full model has a mean Jain value of approximately 0.288, while the no-fairness variant is approximately 0.286. The difference is therefore modest. Similarly, removing priority changes the overall trade-off but does not reverse the main conclusion.

The single-objective configuration provides a more important result. Its mean latency increases to 68.62 ms and QoS satisfaction falls to 78.92%, showing that optimizing a single performance dimension is insufficient for a balanced QoS objective.

3.7 Statistical Testing

A total of 560 paired comparisons were produced across the five-seed experiments. Paired t -tests produced raw $p < 0.05$ in 527 comparisons, and 101 comparisons remained below 0.05 after Holm correction for multiple comparisons.

However, none of the exact two-sided Wilcoxon signed-rank comparisons was significant at the 0.05 level. With only five paired observations, the minimum attainable exact two-sided Wilcoxon p -value is 0.0625. Consequently, the statistical evidence does not establish broad superiority of the proposed blended method.

The difference between the paired t -test and Wilcoxon conclusions also shows why the five-seed design should be interpreted cautiously. The t -test can detect consistent mean differences under its assumptions, while the exact Wilcoxon test is severely limited by the very small number of independent pairs. More repetitions would be needed for stronger statistical power.

The final paper should therefore report the statistical findings as supporting evidence of measurable differences rather than as proof that the predictive allocator is universally superior.

3.8 Discussion

The main research finding is that machine learning can predict future traffic demand substantially better than a persistence assumption, but that prediction accuracy does not automatically translate into better network-level resource allocation.

Random Forest reduces MAE by approximately 43.9% and RMSE by approximately 45.0% relative to persistence and achieves $R^2 = 0.6941$. This establishes that the traffic traces contain useful predictive information.

However, the prediction-only allocation performs worse than the reactive baselines. This is expected when a forecast contains errors and its output is directly converted into resource allocations. A delayed or incorrect demand estimate can allocate bandwidth in anticipation of traffic that does not arrive as predicted while failing to react to short-term fluctuations.

The blended strategy addresses this limitation by combining 75% predicted demand with 25% current demand. It improves substantially over prediction-only allocation. Latency decreases from 28.26 to 16.28 ms, packet loss decreases from 15.72% to 12.82%, jitter decreases from 15.73 to 4.06 ms, and QoS satisfaction increases from 91.15% to 92.18%. Thus, retaining current demand makes the predictive strategy more robust.

Nevertheless, the reactive methods remain stronger on the immediate QoS metrics. Round Robin and Proportional Fair achieve higher throughput, lower latency, lower packet loss and higher QoS satisfaction. Therefore, the proposed method should not be presented as a universal replacement for reactive scheduling.

The advantage of the proposed method is instead the trade-off between fairness and utilization. The blended method obtains Jain fairness of 0.291 and resource utilization of 83.43%, compared with approximately 0.260–0.267 fairness and 78.27–78.29% utilization for the main reactive baselines. This indicates that predictive information can be useful for making resource decisions more demand-aware, although the present allocation mechanism does not yet extract enough benefit to outperform reactive policies on every QoS metric.

This finding also changes how the novelty of the project should be positioned. The contribution is not the claim that ML automatically improves 5G QoS. Rather, the work demonstrates the complete chain from future-demand prediction to resource allocation and shows

the practical trade-off between proactive prediction, immediate QoS, fairness and resource utilization.

There are several limitations. The downloaded traffic representation does not document the bitrate unit or exact wall-clock duration of an observation. The larger scalability scenarios contain synthetic streams derived from 15 real application traces. Queue capacity, allocation floors and delay bounds are simulation assumptions. The evaluation also uses five seeds, which limits statistical power. Finally, the network metrics are simulation outputs rather than measurements from a physical 5G or O-RAN testbed.

These limitations are important because they define the scope of the current findings. Within the implemented simulation framework, the results support the predictive value of Random Forest and the usefulness of blending prediction with current demand, but they do not demonstrate that the proposed allocator is superior to all reactive methods.

3.9 Summary of Findings

The completed experiments lead to four main findings. First, Random Forest provides substantially better future-demand prediction than persistence. Second, a blended demand model with $\lambda = 0.75$ performs better than prediction-only allocation, demonstrating the benefit of retaining current network information. Third, reactive Round Robin and Proportional Fair allocation still provide stronger immediate QoS under the current simulation conditions. Fourth, the proposed predictive blend provides a higher fairness and resource-utilization trade-off, revealing the central research challenge: accurate traffic prediction is useful, but converting predictions into allocation decisions without sacrificing immediate QoS requires a more sophisticated allocation strategy.

4 Conclusion

This study investigated a machine learning-based approach for dynamic QoS optimization and resource allocation in 5G networks under changing traffic conditions. The framework combines traffic-demand prediction with current demand, user priority, QoS urgency, and bounded fairness to determine resource allocation dynamically. A Random Forest regression model was used to predict near-future downlink traffic demand from historical application traffic. The model achieved an MAE of 95,939.54, an RMSE of 275,499.87, and an R^2 of 0.6941, substantially improving over the persistence baseline. These results demonstrate that machine learning can capture useful patterns in traffic behavior and provide meaningful short-term demand estimates.

However, the resource-allocation experiments revealed an important distinction between prediction accuracy and allocation performance. Although the predictive model performed considerably better than persistence, relying heavily on predicted demand did not automatically improve immediate QoS. The prediction-only strategy produced higher latency, packet loss, and jitter than reactive approaches. The selected blended strategy, using 75% predicted demand and 25% current demand, behaved more robustly than prediction-only allocation and achieved higher resource utilization and improved fairness relative to current-demand-only allocation. Nevertheless, Round Robin, Proportional Fair, and reactive demand-based methods achieved better immediate throughput, latency, packet loss, jitter, and QoS satisfaction. Therefore, the main contribution is not a claim of universal superiority, but an empirical demonstration that

future-demand information can influence resource utilization and fairness while introducing a trade-off with immediate QoS.

4.1 Limitations

Several limitations should be considered when interpreting the results. First, the experiments used a Python-based resource-allocation simulator rather than a physical 5G testbed or a full 5G network simulator such as ns-3/5G-LENA. Consequently, the latency, packet-loss, and utilization values represent the behavior of the proposed model under the defined simulation assumptions and should not be interpreted as direct measurements of a deployed 5G system. Second, the available traffic dataset did not provide sufficiently documented wall-clock timing and bitrate units. The experiments therefore treated observations as sequential traffic samples, limiting direct interpretation in real-world time.

Third, the 25-, 50-, and 100-stream experiments use synthetic streams derived from real application profiles. These are controlled scalability scenarios rather than measurements involving physically observed users at those scales. Fourth, the statistical evaluation used five random seeds, which limits statistical power and makes strong significance claims difficult for every metric. Finally, the Random Forest model was trained offline and the framework does not address online model updating, concept drift, inference latency, or the computational overhead of continuous deployment in a 5G RAN.

4.2 Future Work

Future research should evaluate the proposed allocation strategy in ns-3/5G-LENA, O-RAN, or a real experimental 5G platform to determine whether the observed trade-offs persist under realistic radio, scheduling, mobility, and channel conditions. Richer temporal information, including timestamps, mobility, radio-channel measurements, signal quality, and application-level QoS requirements, can also be incorporated. More advanced forecasting models such as LSTM, Temporal Convolutional Networks, Transformers, or lightweight online learning can be investigated.

Another important direction is to formulate allocation as a constrained multi-objective optimization problem that explicitly balances throughput, latency, packet loss, fairness, and resource utilization rather than relying mainly on demand prediction. Adaptive selection of the prediction weight λ based on traffic volatility and network load may further reduce the impact of prediction errors. Finally, larger experiments with more repetitions and additional real-world datasets should be conducted to improve statistical confidence and assess generalization. These extensions can help transform the current predictive resource-allocation framework into a more robust, adaptive, and practically deployable solution for intelligent 5G and beyond-5G networks.

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